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Transformer-based structural connectivity networks for ADHD-related connectivity alterations

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Abstract

Background Attention-deficit/hyperactivity disorder (ADHD) is a common neurodevelopmental disorder that affects behavior, attention, and learning. Current diagnoses rely heavily on subjective assessments, underscoring the need for objective imaging-based methods. This study aims to explore whether structural connectivity networks derived from MRI can reveal alterations associated with ADHD and support data-driven understanding.
Methods We collected brain MRI data from 947 individuals (aged 7–26 years; 590 males, 356 females, 1 unspecified) across eight centers, sourced from the Neuro Bureau ADHD-200 preprocessed dataset. Transformer-based deep learning models were used to learn relationships between different brain regions and construct structural connectivity networks. To prepare input for the model, each region was transformed into a standardized data sequence using four different strategies. The strength of connectivity between brain regions was then measured to identify structural differences related to ADHD. Five-fold cross-validation and statistical analyses were used to evaluate model robustness and group differences, respectively.
Results Here we show that the proposed method performs well in distinguishing ADHD individuals from healthy controls, with accuracy reaching 71.9 percent and an area under curve of 0.74. The structural networks also reveal significant differences in connectivity patterns (paired t-test: $P = 0.81 \times 10^{-6}$), particularly involving regions responsible for motor and executive function. Notably, the importance rankings of several brain regions, including the thalamus and caudate, differ markedly between groups.
Conclusions This study shows that ADHD may be associated with connectivity alterations in multiple brain regions. Our findings suggest that brain structural connectivity networks built using Transformer-based methods offer a promising tool for both diagnosis and further research into brain structure.

Plain Language Summary

Attention-deficit/hyperactivity disorder (ADHD) is a brain condition that affects attention, behavior and learning, especially in children. Diagnosing ADHD is currently based on interviews and observations, which can be subjective and vary between doctors. This study explored whether artificial intelligence models could detect brain structure differences in people with ADHD using MRI scans. Researchers used a deep learning method called a Transformer to analyze brain images from 947 individuals. The model was able to spot patterns that helped distinguish people with ADHD from those without. It also highlighted brain areas that may be especially important in ADHD. These findings suggest that artificial intelligence could help build more accurate and objective tools for ADHD diagnosis, supporting better understanding and care in the future.

Attention-deficit hyperactivity disorder (ADHD) stands as the most prevalent neurodevelopmental disorder, with its incidence steadily increasing over the years[1]. In the United States alone, between 2016 and 2019, ~9.8% of the population aged 3–17 years, totaling nearly 6,000,000 children and adolescents, were diagnosed with ADHD. ADHD not only impacts the health and quality of life of affected individuals, but also increases the risk of other psychiatric disorders[2]. Early detection and intervention can markedly alleviate symptoms, enhance attention, and facilitate a return to normalcy in

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learning and daily life[3]. ADHD is influenced by a range of genetic, environmental, and psychosocial factors, including family history, perinatal complications, and early-life stress[4]. Despite these known risk factors, diagnosis still relies heavily on clinical interviews and subjective assessments[5], underscoring the need for more objective and accessible approaches.

Several neuroimaging studies have explored the potential biomarkers and structural alterations associated with ADHD[6]. Findings suggest that ADHD is linked to abnormalities in gray matter volume, cortical thickness, and white matter integrity, particularly in regions involved in attention control, impulse regulation, and executive function (e.g., the prefrontal cortex, basal ganglia, and cerebellum). Recent studies have also suggested that structural connectivity disruptions play a key role in ADHD pathophysiology[7–10]. Diffusion tensor imaging (DTI) studies have reported altered white matter connectivity patterns[11], suggesting that ADHD may involve distributed network-level alterations rather than localized structural deficits. Understanding these connectivity patterns could provide insights into the neuropathological basis of ADHD and facilitate the identification of objective neurobiological markers relevant to ADHD diagnosis[12].

Three-dimensional (3D) MRI T1-weighted imaging (T1WI) yields high-resolution anatomical brain structure images, offering valuable insights into ADHD-related structural changes. Artificial intelligence (AI) has been increasingly utilized to enhance the automatic quantification of structural changes in 3D T1WI images for the diagnosis of ADHD[13–15]. This approach leverages advanced algorithms, such as 3D-convolutional neural networks (3D-CNN), to analyze complex structural alterations in the brain associated with ADHD, achieving a diagnostic accuracy of ~70%. However, current AI approaches are limited to analyzing isolated regional abnormalities[13–15], overlooking distributed changes in brain connectivity. Some studies have manually constructed structural brain networks using T1WI images[10], revealing that coordinated changes across multiple brain regions are associated with ADHD. These networks are predefined and lack the ability to be automatically derived from data. This highlights the need for a learnable, data-driven network-based approach to comprehensively investigate ADHD-related structural alterations.

The Transformer network incorporates a self-attention mechanism capable of identifying interrelationships between elements. Recent advancements have seen the emergence of the vision transformer model, which is designed to process images directly, surpassing traditional CNN models in image classification tasks[16]. Within the realm of neuroimaging, several studies have leveraged the interrelated characteristics of the Transformer's self-attention module to construct functional connectivity networks. These networks are established by calculating correlations between fMRI time series, thereby enhancing diagnostic performance in neurological diseases[17–20]. Consequently, Transformer holds promise for mining interrelationships between different brain regions, facilitating the identification of connections and dependency information among these regions.

In this study, we propose a method for constructing brain structural connectivity networks (SCNs) from 3D T1WI images to enhance the understanding of ADHD-related neuroanatomical alterations. By leveraging the self-attention mechanism of the Transformer, structural correlations among different brain regions can be learned to improve the diagnostic accuracy of ADHD. The approach of this study involves mapping MRI images to an Automated Anatomical Labeling (AAL) template and processing the images of each brain region into sequences of uniform length, serving as input for the self-attention module of the Transformer encoder. Four models are constructed based on different processing modes: a feature-Transformer (F-T) model and three voxel-based Transformer models. We evaluate the diagnostic performance of the models using the ADHD-200 public dataset[21], and analyze the structural connectivity patterns of the SCN to explore the clinical application value of the proposed model. The proposed method effectively distinguishes individuals with ADHD from healthy controls, achieving 71.9% accuracy and an AUC of 0.74. Structural connectivity networks reveal significant differences, particularly in motor and executive function regions, with notable variations in the importance of areas such as the lingual gyrus, precuneus lobe, thalamus and caudate. These findings suggest that Transformer-based structural connectivity networks hold promise for both ADHD diagnosis and further research into its neural mechanisms.

Methods

Data description

This study utilized data from the publicly available ADHD-200 dataset, which was accessed through the Neuroimaging Informatics Tools and Resources Clearinghouse (NITRC) at https://fcon_1000.projects.nitrc.org/indi/adhd200/, in accordance with the policies of the 1000 Functional Connectomes Project and the International Neuroimaging Data-Sharing Initiative (INDI). The dataset comprises 776 samples in the training set and 197 samples in the testing set collected from eight centers: Peking University (PKU), Kennedy Krieger Institute (KKI), NeuroIMAGE Sample (NeuroIMAGE), New York University Child Study Center (NYU), Oregon Health and Science University (OHSU), University of Pittsburgh (UPittsburgh), Washington University in St. Louis (WashU), and Brown University. Data were used solely for non-commercial research purposes. Access to the dataset required free registration with NITRC and agreement to the terms of the ADHD-200 data usage license. All imaging and phenotypic data in the ADHD-200 dataset were fully anonymized and obtained under Institutional Review Board (IRB) approvals at the respective contributing institutions. According to INDI protocol, no additional IRB approval was required for this secondary analysis of de-identified, publicly available data. Informed consent was obtained by the original data providers and waived for this study. The diagnostic criteria for all the samples are publicly available. The 26 testing samples from Brown University were excluded from the study because of unavailable diagnostic results. Overall, the dataset comprises 362 patients with ADHD and 585 HCs, and the clinical information is summarized in Table 1.

The preprocessed 3D T1WI images, obtained from the Neuro Bureau ADHD-200 preprocessed database, were downloaded based on the Athena pipeline[21] The image preprocessing methods included: First, the MRI T1WI images were reoriented into right posterior inferior (RPI) orientation using the '3dresample' tool of AFNI; Second, the skulls on the images were stripped using the '3dSkullStrip' tool of AFNI; Third, an initial linear transformation was computed between the anatomical and template images using the 'flirt' tool of FSL; Fourth, the transformation was refined using the non-linear transformation in the 'fnirt' tool of FSL; Last, the skull-stripped anatomical images were written into the template space at 1 mm × 1 mm × 1 mm resolution using the 'applywarp' tool of FSL. The images used in this study were skull-stripped T1WI images with a voxel size of 1 mm × 1 mm × 1 mm and an image matrix of 189 × 233 × 197. To standardize the data for subsequent analysis, SimpleITK was employed to resample the images to a resolution of 1.5 mm × 1.5 mm×1.5 mm, resulting in a processed image matrix of 121 × 145 × 121. The brain parcellation scheme utilized in this study is the AAL template, which partitions the entire brain into 116 regions, including 90 cerebral regions and 26 cerebellar regions[22]. The AAL template is widely used in neuroimaging studies[10,18,23] and provides a well-established, standardized parcellation scheme that facilitates comparison with prior literature.

Model construction

Four models were developed: the feature-based F-T model, the voxel-based voxel random sampling-Transformer (VRS-T) model, the voxel-based original voxel embedding-Transformer (OVE-T) model, and the voxel-based area-wise image resampling-Transformer (AIR-T) model (Fig. 1a, b). The central component of these models is the Transformer encoder, comprising a multi-head attention (MHA) module, normalization layers, and a feed-forward network (FFN) (Fig. 1b). The MHA module computes the interrelationships among the input sequences and generates attention maps. Additionally, we investigated whether the integration of the four models can improve performance by developing two fusion models (Supplementary Fig. 1).

Table. 1 | Clinical information of subjects in the ADHD-200 dataset (n = 947)

Center	ADHD-200 training set			ADHD-200 testing set		
	Subjects (ADHD/HC)	Age (years) (mean ± SD)	Gender (male/female)	Subjects (ADHD/HC)	Age (years) (mean ± SD)	Gender (male/female)
PKU	194 (78/116)	11.98 ± 1.86	142/52	51 (24/27)	10.64 ± 1.94	32/19
KKI	83 (22/61)	10.24 ± 1.34	46/37	11 (3/8)	10.06 ± 1.28	10/1
NeuroIMAGE	48 (25/23)	16.99 ± 2.71	31/17	25 (11/14)	18.87 ± 3.22	12/13
NYU	222 (123/99)	11.59 ± 2.93	142/79	41 (29/12)	10.73 ± 2.69	28/13
OHSU	79 (37/42)	8.83 ± 1.12	44/35	34 (6/28)	9.71 ± 1.31	17/17
UPittsburgh	89 (0/89)	15.11 ± 2.88	46/43	9 (4/5)	14.82 ± 1.06	7/2
WashU	61 (0/61)	11.47 ± 3.85	33/28	0	/	/
Overall	776 (285/491)	11.99 ± 3.21	484/291	171 (77/94)	11.86 ± 3.79	106/65

ADHD attention-deficit hyperactivity disorder, *HC* healthy control, *KKI* Kennedy Krieger Institute, *NYU* New York University, *OHSU* Oregon Health and Science University, *PKU* Peking University, *SD* standard deviation, *UPittsburgh* University of Pittsburgh, *WashU* Washington University.

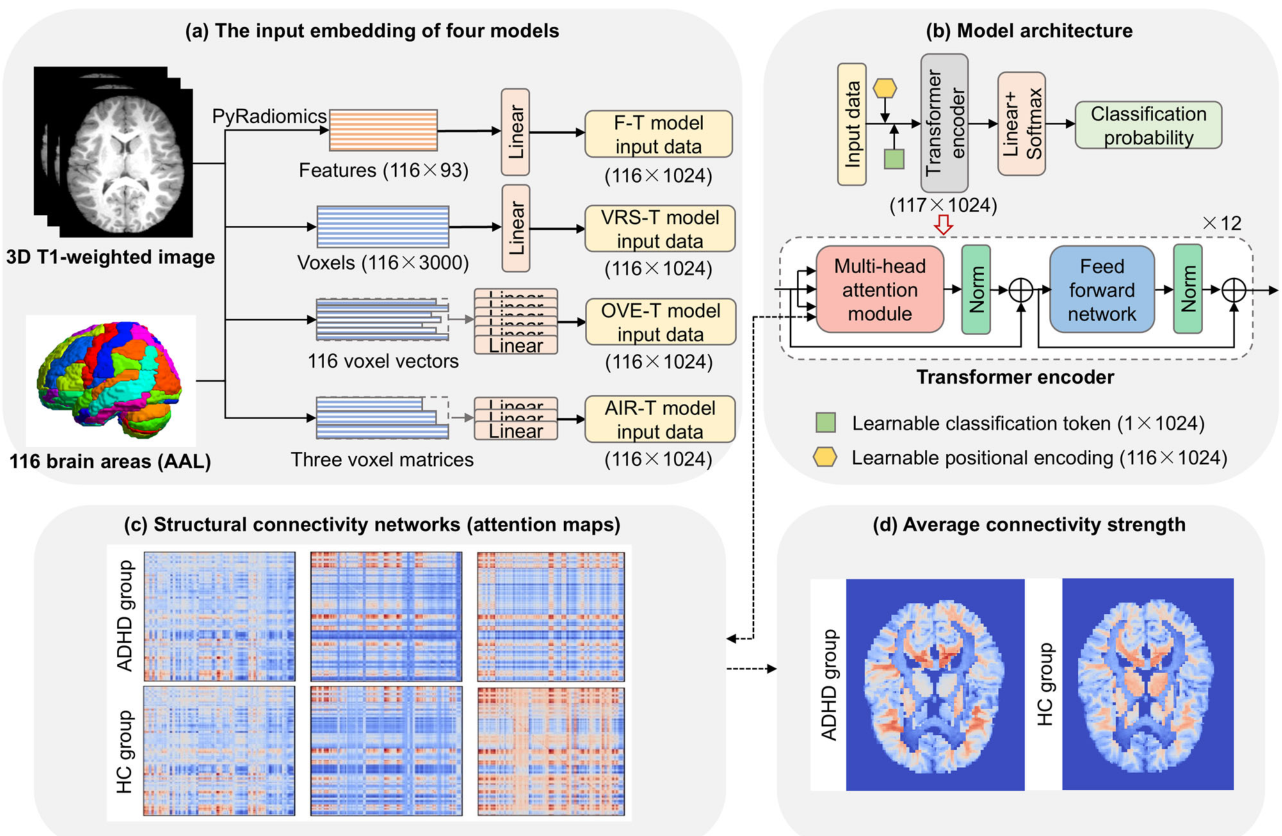

Fig. 1 | Overview of model construction and structural connectivity network derivation. a Four types of image processing methods were utilized to transform the images within each brain region into the input of the Transformer encoder, thereby facilitating the construction of the four models. **b** Architecture of the four models. **c** The attention maps derived from the models corresponded to the structural connectivity networks identified in this study. **d** The structural connectivity networks were quantified using the average connectivity strength to uncover ADHD-related alterations in brain structure.

The F-T model initially transformed images from each brain region into radiomics features, which showed superior performance in ADHD diagnosis over traditional morphological features[24], using the Pyradiomics tool (Fig. 1a). A total of 93 features were extracted from each of the 116 brain regions, yielding an input feature matrix $F \in \mathbb{R}^{116\times93}$ for the model. Among the 93 radiomics features, 18 were first-order features, while the remaining 75 were higher-order texture features. Among the texture features, 24 were derived from the gray-level co-occurrence matrix (GLCM), 16 from the gray-level run length matrix (GLRLM), 16 from the gray level size zone matrix (GLSZM), 14 from the gray level dependence matrix (GLDM), and 5 from the neighboring gray tone difference matrix (NGTDM).

The linear projection of the feature matrix F was added with a randomly initialized positional encoding matrix P. Subsequently, it was concatenated with a randomly initialized classification token C to create the input matrix for the Transformer encoder (Fig. 1b). During model training, the values of P and C were automatically adjusted to optimize the performance. The Transformer encoder comprised 12 layers, with the MHA module serving as its core component. The MHA module facilitates parallel self-attention heads, enabling diverse attention mechanisms and allowing the concentration of attention in different brain regions[25]. The self-attention mechanism in the MHA module is calculated as follows:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q represents the query matrix; K denotes the key matrix; and V signifies the value matrix. Notably, the attention map QK^T in Eq. (1), excluding the classification token C, computes the interrelations among the 116 brain regions, referred to as the SCN in this study.

The output matrix of the final encoder layer synthesized global features with the aid of the classification token C, yielding 1×1024-dimensional classification information. This output was then projected through a linear layer to derive the binary classification scores. Finally, the Softmax function was applied to convert these scores into probabilities indicating the likelihood of the subject belonging to either the ADHD or HC group.

The voxel count varies within the 116 brain regions as delineated by the AAL template, precluding a straightforward input into the Transformer encoder. Consequently, this study introduced three voxel models aimed at standardizing voxel sequences across brain regions to suit the requirements of the Transformer encoder.

The VRS-T model uniformly sampled a fixed number of voxels from each brain region, aggregating these voxel sequences into a voxel matrix, which was subsequently processed and fed into the Transformer encoder (Supplementary Fig. 2a). Specifically, in our study, the number of voxels within each of 116 brain regions ranged from 88 to 12,080 (median: 3184). In the VRS-T model, we sampled 3000 voxels from each brain region to preserve as much original information as possible for large brain regions while minimizing interpolation in small brain regions. The training set was randomly sampled to enhance the robustness of the model, while the testing set was sampled at regular intervals to keep the stability of the results. The voxels of 116 brain regions were concatenated to compose a voxel matrix $V_{VRS} \in \mathbb{R}^{116 \times 3000}$, whose linear projection was added with a learnable positional encoding matrix P and then concatenated with a learnable classification token C as the input of the Transformer encoder. The values of P and C were automatically learned during the model training process. The main architecture and hyper-parameters of the VRS-T model were the same as those of the F-T model.

The OVE-T model projected the original voxels from each brain region onto a common dimension. The transformed voxel sequences from all brain regions were then amalgamated into a voxel matrix, which served as the input for the Transformer encoder (Supplementary Fig. 2b). Specifically, the OVE-T model transformed all $1 \times n$ voxels of each brain region into a 1×1024 vectors using a linear projection layer and then concatenated the transformed vectors of 116 brain regions into a voxel matrix $V_{OVE} \in \mathbb{R}^{116 \times 1024}$. V_{OVE} was added with a learnable positional encoding matrix P and then concatenated with a learnable classification token C as the input of the Transformer encoder. The values of P and C were automatically learned during the model training process. The main architecture and hyperparameters of the OVE-T model were the same as those of the F-T model.

The AIR-T model partitioned T1WI scans of different brain regions, resizing them to one of three standardized image sizes based on their initial dimensions. The resized images were converted into three voxel matrices and linearly projected onto the same dimension. The resulting linear projections were then combined into a voxel matrix suitable for input into the Transformer encoder (Supplementary Fig. 2c). Specifically, the sizes of 116 brain regions vary largely on MRI T1WI images, with bounding box ranging from $4 \times 4 \times 8$ to $30 \times 72 \times 24$. The large differences between brain regions made it difficult to simply resample images of all brain regions into the same size. A small resampling size may cause the loss of most information of large brain regions while a large resampling size may lead to poor resampling results. Therefore, AIR-T model categorized 116 brain regions into three types according to their original size and resampled them into three types of image sizes. The 52 brain regions with maximum 3D length smaller than 30 were resampled to the size of $10 \times 10 \times 10$, resulting in 1000 voxels of each resampled brain region; the 49 brain regions with maximum 3D length between 30 and 50 were resampled to the size of $15 \times 15 \times 15$, resulting in 3375 voxels of each resampled brain region; the 15 brain regions with maximum 3D length larger than 50 were resampled to the size of $20 \times 20 \times 20$, resulting in 8000 voxels of each resampled brain region. Before resampling, we segmented each brain region from original images and generated a cube image using the maximum 3D length with the regions beyond the brain region were filled with 0. The voxels of 116 resampled brain regions were flattened and concatenated to three types of voxel matrices: $V_{AIR1} \in \mathbb{R}^{52 \times 1000}$, $V_{AIR2} \in \mathbb{R}^{49 \times 3375}$, $V_{AIR3} \in \mathbb{R}^{15 \times 8000}$. The linear projection of the three matrices were concatenated to a voxel matrix $V_{AIR} \in \mathbb{R}^{116 \times 1024}$, which was added with a learnable positional encoding matrix P and then concatenated with a learnable classification token C as the input of the Transformer encoder. The values of P and C were automatically learned during the model training process. The main architecture and hyperparameters of the AIR-T model were the same as those of the F-T model.

Two fusion models were designed, namely a strategy fusion model and an information fusion model, with the aim of integrating the strengths of all single models to enhance diagnosis performance. The strategy fusion model combined the majority voting method and the average probability method to merge the classification results and output probabilities of the four single models (Supplementary Fig. 1c). Specifically, the mean output probabilities of the four single models were computed and a voting mechanism was employed for subject diagnoses. If none of the models diagnosed the subject as ADHD, the final diagnostic result was HC. Conversely, if three or more models indicated ADHD for a subject, the diagnostic outcome was ADHD. In cases where only one or two models suggested ADHD, the final diagnosis was determined based on the class with the higher mean probability. The information fusion model utilized a feature-guided co-attention module to fuse and refine the classification information from the four single models (Supplementary Fig. 1d). Within the co-attention module, the classification token C of the F-T model served as the Q matrix, while the classification tokens C of the three voxel-based models acted as the K and V matrices, respectively. MHA was computed according to Eq. (1). To streamline training and maintain computational efficiency, the classification tokens C from the four trained single models were extracted, output, and subsequently retrained within the co-attention Transformer network. This approach effectively integrated classification information from all four models, leveraging feature guidance to refine voxel information and enhance diagnosis performance.

Model implementation

To ensure consistency and comparability with the ADHD-200 Global Competition, we adopted the pre-defined ADHD-200 training and testing sets. The pre-defined ADHD-200 training set was further split into the training and validation subsets at a ratio of 4:1. The training subset was used to train the model; the validation subset was employed to monitor the performance of the model during training and to select the optimal hyperparameters; the pre-defined ADHD-200 testing set, which was not seen during model training and selection, was only used to test the performance of the trained model.

This study employed accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and balanced accuracy as the evaluation metrics to assess the diagnostic performance of the models. Balanced accuracy, defined as the arithmetic mean of sensitivity and specificity, concurrently evaluates the comprehensive diagnostic performance of the model for both the ADHD and HC classes.

This study was conducted on a server equipped with an Intel(R) Xeon(R) Silver 4210 CPU (2.20 GHz) and NVIDIA GeForce RTX 2080 Ti GPU (CUDA version: 10.2) using PyTorch 1.9.0 framework. The training hyperparameters of the models were as follows: (1) Number of Transformer encoder layers = 12; (2) Number of heads in MHA = 8; (3) Optimization:

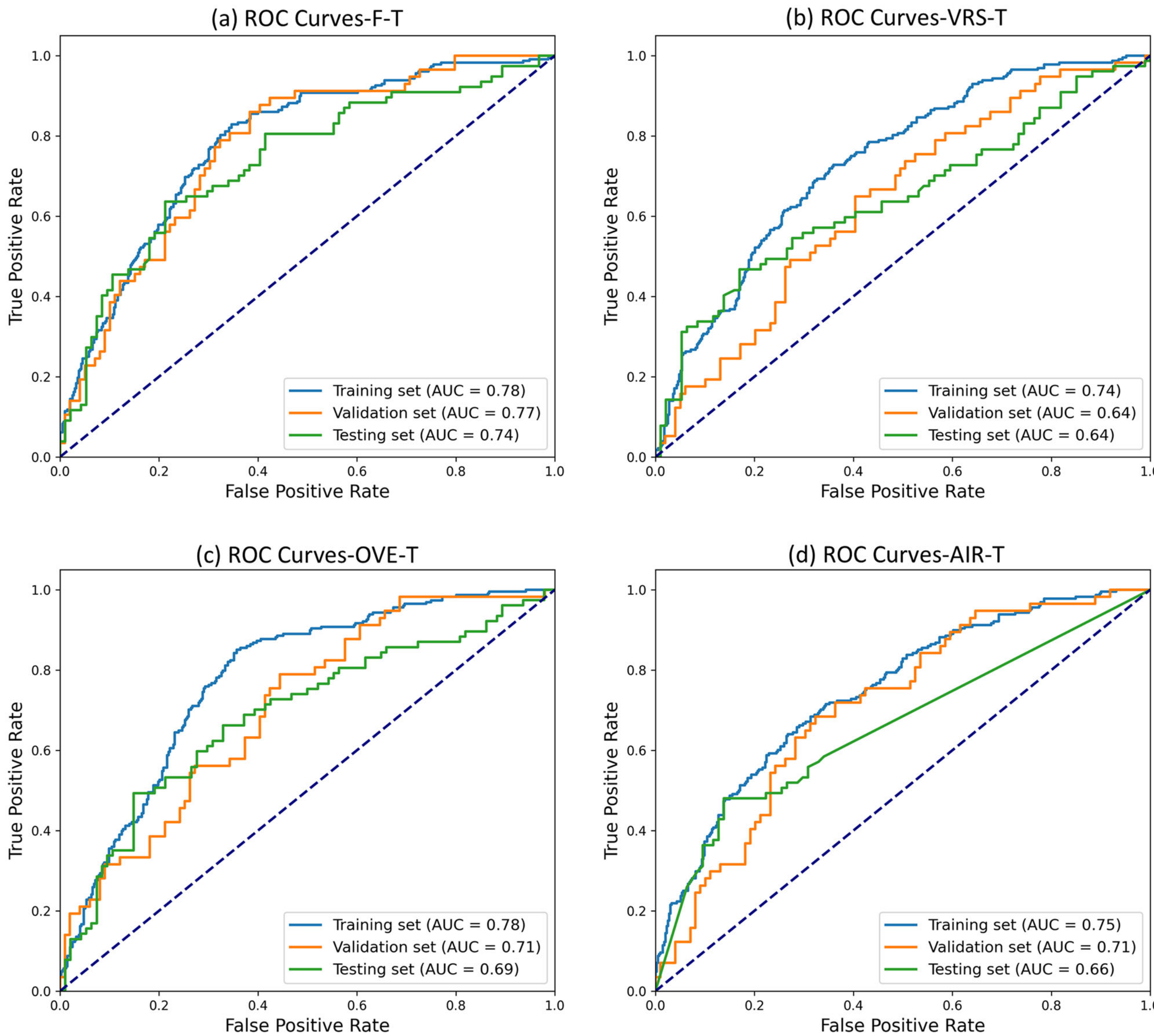

Fig. 2 | Receiver operating characteristic (ROC) curves of the four models on the ADHD-200 dataset. a ROC curves of the feature-Transformer (F-T) model. **b** ROC curves of the voxel random sampling-Transformer (VRS-T) model. **c** ROC curves of the original voxel embedding-Transformer (OVE-T) model. **d** ROC curves of the area-wise image resampling-Transformer (AIR-T) model. For all models, the training, validation, and testing sets included 620, 156, and 171 participants, respectively.

RMSProp; (4) Loss function: cross-entropy; (5) Learning rate: 1×10^{-3}; (6) Batch size: 1; (7) Epoch = 300; (8) Dropout = 0.1.

SCN visualization

The SCN visualization was performed using the Python package 'Visualizer' (https://github.com/luo3300612/Visualizer). Specifically, we first used the 'get_local' function from 'Visualizer' to obtain 8×12 (heads × layers) attention maps for each subject. Each attention map was then rescaled using the softmax function. Next, the attention maps from all heads were averaged for each subject, reducing the number of attention maps to 12 (one per layer). For each layer, the attention maps of all subjects in each group (ADHD/HC) were averaged to generate one group-level attention map, resulting in 12 attention maps per group. The average of these 12 encoder layers was used as the SCN value for each group. We further calculated symmetric SCN values by averaging the SCN values in both directions for each pair of brain regions. Finally, the attention maps were visualized using the 'matplotlib.pyplot' package.

Robustness experiments

We conducted five-fold cross-validation to assess the robustness of the performance of the proposed models. Specifically, 947 samples in the ADHD-200 dataset were split into five subsets. In each fold, one subset served as the testing set to evaluate the model performance, whereas the remaining four subsets were combined and split into the training and validation sets at a ratio of 4:1 for model training and selection.

To examine whether the choice of the brain parcellation template affects model performance, we replicated the analysis using the Brainnetome template (http://www.brainnetome.org/), which defines 246 brain regions.

Given that many children with ADHD have at least one comorbid psychiatric disorder, we further investigated the impact of comorbidities on model performance. We divided the ADHD-200 dataset into two cohorts: a non-comorbid cohort (including data from PKU, KKI, OHSU, and WashU), and a comorbid cohort (including data from NeuroIMAGE, NYU, and UPittsburgh). We then conducted an additional analysis by training two separate F-T models: (1) the non-comorbid model, trained and validated using the non-comorbid cohort from the pre-defined ADHD-200 training set, which included only healthy controls and ADHD participants without comorbidities, and (2) the comorbid model, trained and validated using the comorbid cohort from the pre-defined ADHD-200 training set, which also included some ADHD participants with comorbid conditions. Both models were subsequently tested in both cohorts on the pre-defined ADHD-200 testing set.

Statistics and reproducibility

In this study, we analyzed brain MRI T1WI data from 947 individuals across eight centers, obtained from the Neuro Bureau ADHD-200 preprocessed dataset. To evaluate the structural connectivity differences related to ADHD, Transformer-based deep learning models were constructed using standardized regional input sequences derived from four distinct preprocessing strategies. The robustness of these models was assessed using five-fold cross-validation, ensuring that each subject served as a test sample exactly once. All model training procedures and statistical analyses were performed using reproducible scripts developed in Python and PyTorch. Specific hyperparameters, such as learning rate, batch size, and model architecture details, were held constant across all experiments and are documented in the Methods section. For group comparisons, the names of the statistical tests and corresponding effect size estimates were reported, with sample size fixed by the availability of the public dataset. No data was excluded from analysis. The complete source codes and source data will be made available in a public repository upon publication to ensure full reproducibility of the study.

Results

Model performance

Figure 2 illustrates the receiver operating characteristic (ROC) curves and area under curves (AUCs) of the four models on the ADHD-200 dataset (Supplementary Data 1). The F-T model (AUCs: 0.74) performed better than the three voxel-based models (AUCs: 0.64–0.69) on the ADHD-200 testing set. Supplementary Data 2 shows the performance of the four models in the ADHD-200 testing set. Among the four models, the feature-based F-T model exhibited a significantly superior overall sensitivity (63.6%) compared to the three voxel-based models (32.5%, 46.8%, 48.1%; McNemar's test: $P = 3.03 \times 10^{-6}$–0.05, estimated odds ratio 2.20 [1.04, 4.65]–13.00 [3.09, 54.77]). While the accuracy and balanced accuracy of the three voxel models ranged from 66.1% to 69.0% and from 63.1% to 67.1%, respectively, the F-T model demonstrated enhancements in both accuracy (71.9%, McNemar's test: $P = 0.19$–0.60, estimated odds ratio 1.19 [0.71, 2.01]–1.53 [0.86, 2.72]) and balanced accuracy (71.2%).

As shown in Supplementary Data 2, the feature-based F-T model exhibited superior performance on the PKU, KKI, NeuroIMAGE, and OHSU datasets, achieving accuracies of 68.6%, 90.9%, 84.0%, and 70.6%, sensitivities of 33.3%, 66.7%, 81.8%, and 66.7%, and balanced accuracies of 66.7%, 83.3%, 83.8%, and 69.1%, respectively. In contrast, the three voxel-based models showed lower performance, with accuracies ranging from 52.9% to 58.8%, 72.7% to 81.8%, 56.0% to 84.0%, and 82.4%, sensitivities ranging from 0.0% to 25.0%, 0.0% to 33.3%, 0.0% to 63.6%, and 0.0% to 33.3%, and balanced accuracies ranging from 50.0% to 56.5%, 50.0% to 66.7%, 50.0% to 81.8%, and 50.0% to 63.1%, respectively. Although the VRS-T model showed improved performance on the NYU dataset (balanced accuracy: 68.1%) compared to other single models (balanced accuracy: 56.6% to 63.7%), its sensitivity remained at 0.0% in the other five centers. Notably, the OVE-T model performed optimally on the UPittsburgh dataset, achieving a balanced accuracy of only 52.5%.

The information fusion model achieved the highest overall accuracy, overall sensitivity and overall balanced accuracy (73.7%, 67.5% and 73.1%, respectively) among all models. The strategy fusion model showed higher overall accuracy (73.1%) than single models, but its overall sensitivity (50.6%) and overall balanced accuracy (71.1%) was lower than that of the F-T model. McNemar's test also revealed no significant statistical difference in overall accuracy between the F-T model and the two fusion models ($P = 0.72$ and 0.87, estimated odds ratio 0.82 [0.41, 1.67] and 0.90 [0.48, 1.70]). The overall sensitivity of the information fusion model showed no significant difference with the F-T model ($P = 0.65$, estimated odds ratio 0.73 [0.29, 1.81]). This outcome may be attributed to the fact that the fusion models were still guided by the probabilities and SCN information generated by the F-T model, while incorporating probabilities and information from the three voxel models.

Robustness experiments

In five-fold cross validation, the AUCs of the F-T model were 0.75–0.78, higher than those of the AIR-T (0.62–0.71), OVE-T (0.62–0.71), and VRS-T (0.64–0.75) models. The results of the four models in five-fold cross validation are shown in Fig. 3 and Supplementary Figs. 3–5.

Based on Brainnetome template, the F-T model achieved AUCs of 0.69–0.76 in five-fold cross validation, while the AUCs of the AIR-T, OVE-T, and VRS-T models were 0.63–0.73, 0.63–0.75, and 0.64–0.70. The results of the four models based on Brainnetome template are shown in Supplementary Fig. 6–9.

Results of comorbidities-related experiment showed that the non-comorbid model achieved AUCs of 0.72 on the non-comorbid cohort and 0.68 on the comorbid cohort, while the comorbid model yielded AUCs of 0.73 and 0.72 on the non-comorbid and comorbid cohorts, respectively (Supplementary Fig. 10). No statistically significant differences were observed between the two cohorts for either model (Delong test: $P = 0.69$, estimated effect size 0.04 [−0.13, 0.20] for the non-comorbid model; $P = 0.94$, estimated effect size 0.58×10^{-2} [−0.15, 0.16] for the comorbid model).

SCN visualization

Visualization of the constructed SCNs was conducted to elucidate the diagnostic rationale underlying the models proposed in this study. While the fusion models integrated diagnostic information from the single models, they still relied on the SCNs established by the single models for diagnosis. Hence, the F-T model was utilized, which demonstrated the highest diagnostic performance among all single models, for SCN visualization. Figure 4 illustrates the SCNs, symmetric SCNs, and binary symmetric SCNs at different thresholds of the ADHD and HC groups of the F-T model. The SCNs at various encoder layers are shown in Supplementary Fig. 11. SCNs tend to exhibit modular characteristics and small world properties. Clear disparities are evident in both the pattern (the threshold of 0.6, Fig. 4c) and strength (the threshold of 0.7, Fig. 4d) of SCNs between the ADHD and HC groups, with these differences intensifying as the network layers deepen (Supplementary Fig. 11). ADHD may induce alterations in both the structural connectivity strength and patterns among different brain regions.

To visually illustrate the role of the model in assisting the diagnosis of ADHD, analysis and visualization of the diagnostic importance of each brain region in the SCN was conducted using its average connectivity strength (ACS). ACS is defined as the average of the structural connectivity strengths with all brain regions. Specifically, the 12-layer attention maps of each group in the F-T model were averaged to obtain one average attention map for each group of ADHD and HC. Subsequently, these attention maps were standardized to the range of [0, 1]. After standardization, the row and column of each brain region were summed and averaged to determine its total connectivity strength with all brain regions. This value was then divided by 116 to calculate the ACS. Figure 5 presents the ACS at representative sections in the ADHD and HC groups. In the visualization, redder colors represent stronger strengths and greater importance. Comparison between the HC and ADHD groups reveals several noteworthy differences. Multiple brain regions, such as the lingual gyrus, precuneus lobe, postcentral gyrus, precentral gyrus, and superior parietal gyrus, exhibit increased ACS in the ADHD group. Conversely, in the caudate nucleus and thalamus, the ACSs of the HC group surpass those of the ADHD group.

Significant differences in the ACSs were observed between the ADHD and HC groups (0.27–0.66 vs. 0.28–0.51; paired t-test: $P = 0.81 \times 10^{-6}$, Cohen's d estimate 0.40 [0.24–0.56]). Table 2 shows

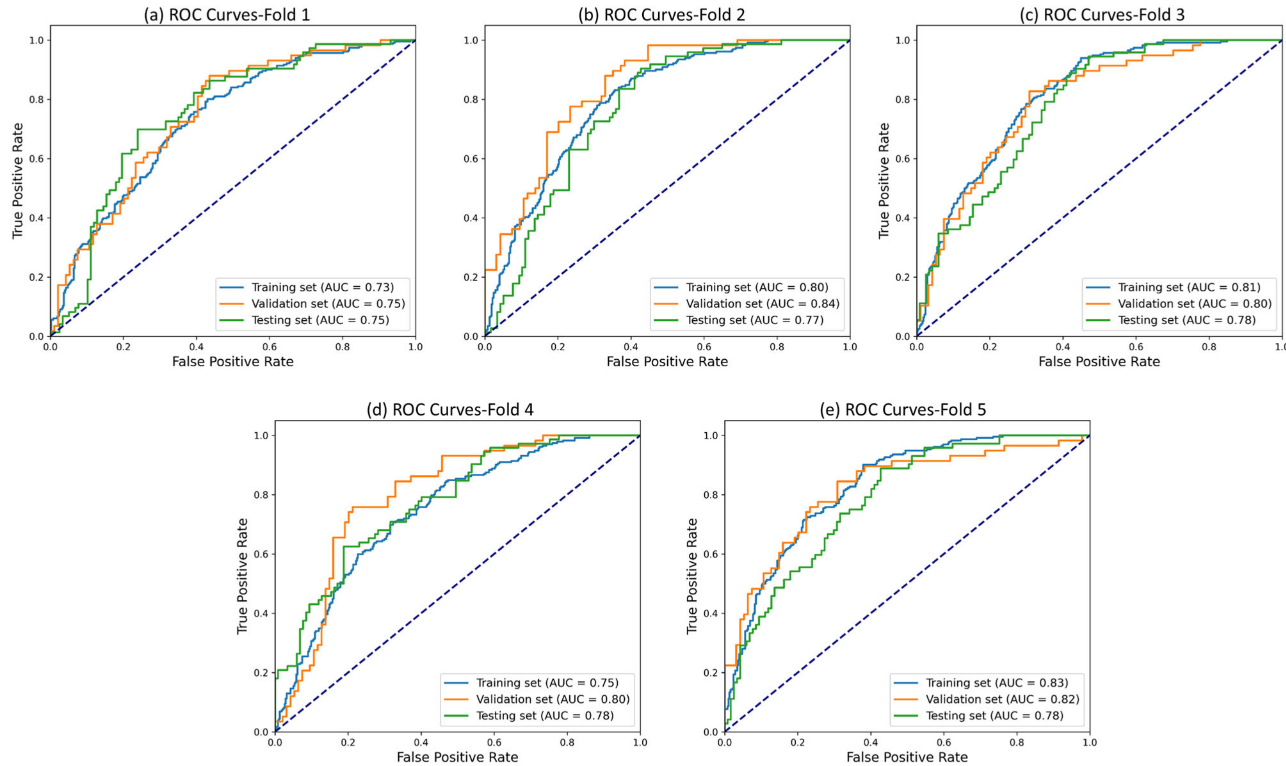

Fig. 3 | Receiver operating characteristic (ROC) curves of the feature-Transformer (F-T) model in five-fold cross validation. Panels (**a**–**e**) show the ROC curves for Folds 1–5, respectively. In Folds 1 (**a**) and 2 (**b**), the training, validation, and testing sets included 605, 152, and 190 participants, respectively. In Folds 3 to 5 (**c**–**e**), the training, validation, and testing sets included 606, 152, and 189 participants, respectively.

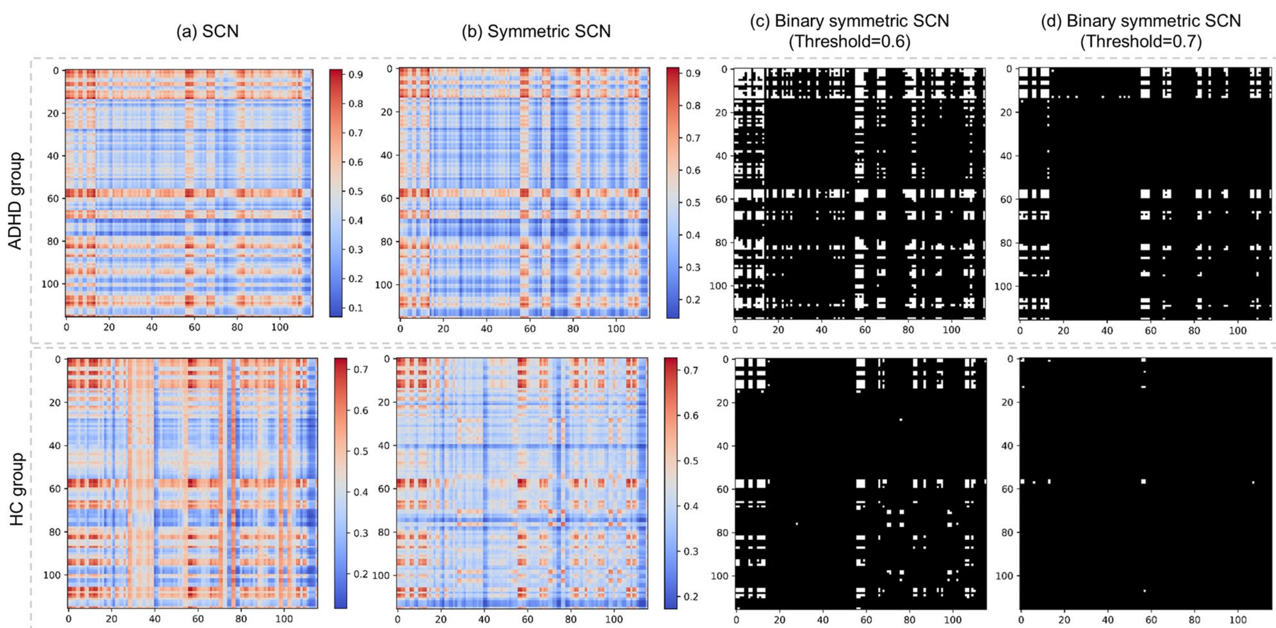

Fig. 4 | Structural connectivity networks (SCNs) of the ADHD and HC groups in the F-T model. a SCNs, the SCN values are the average across all patients and all encoder layers; **b** Symmetric SCNs, the symmetric SCN values of each pair of brain regions are the average of two calculation directions; **c** Binary symmetric SCNs at the threshold of 0.6 for showing the differences in pattern; **d** Binary symmetric SCNs at the threshold of 0.7 for showing the differences in strength. Each row and column in the figures correspond to one of 116 brain regions, while the color bar in (**a**) and (**b**) denotes the strength of structural connectivity between brain regions, with redder hues indicating greater connectivity strength.

the top 10 most important brain regions in the SCNs of both groups, ranked in descending order of ACSs. The bilateral precentral gyrus, left middle frontal gyrus, right triangular part of the inferior frontal gyrus, and bilateral postcentral gyrus emerged as dominant regions in both groups, with their ACSs differing significantly (paired t-test: $P = 0.13 \times 10^{-4}$, Cohen's d estimate 7.78 [2.09–13.48]), highlighting them as key regions for ADHD diagnosis. In the ADHD group, the lingual gyrus showed a higher relative importance, ranking 38th and

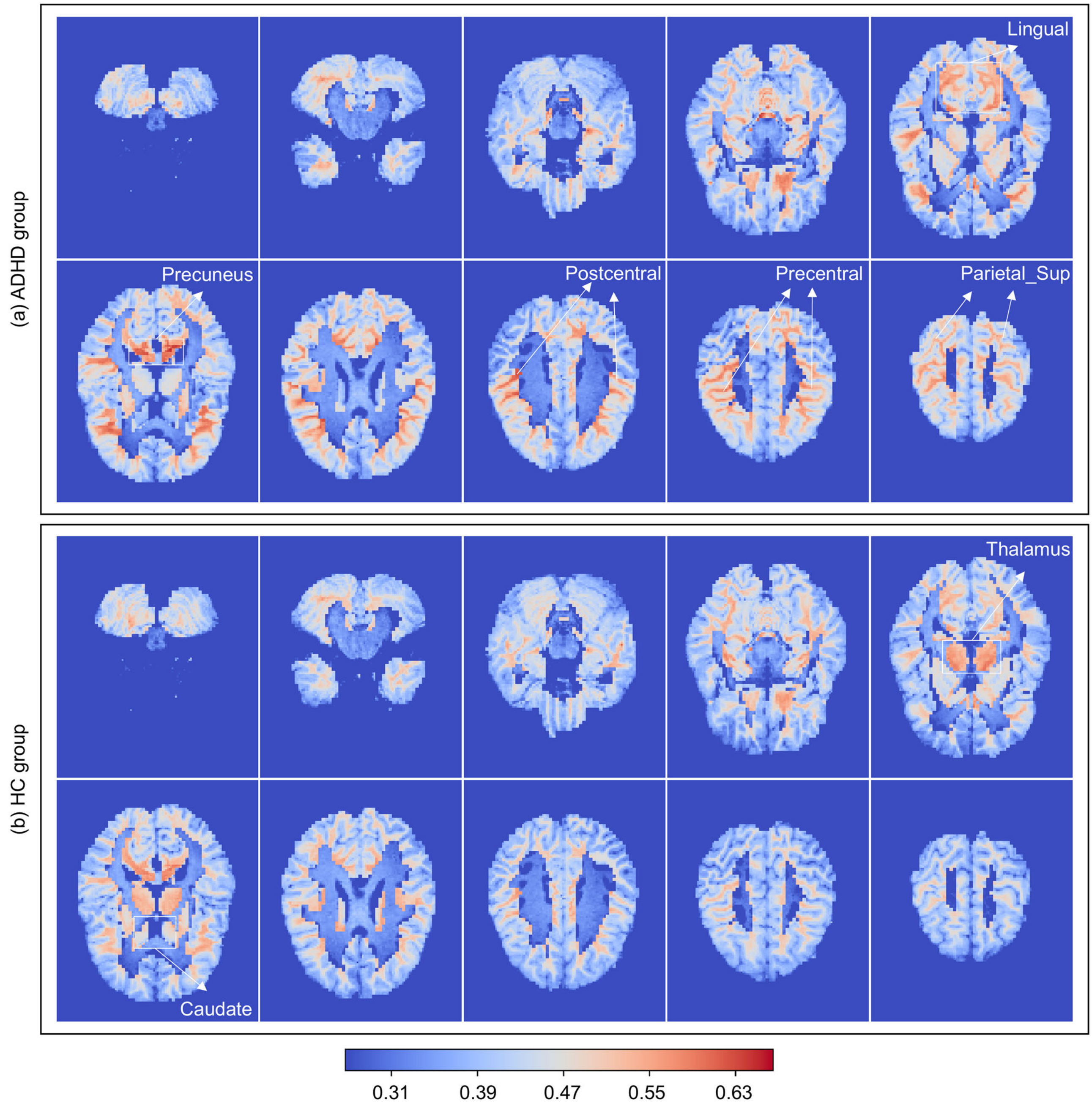

Fig. 5 | Average connectivity strength in representative sections of the feature-Transformer (F-T) model. The average connectivity strength reflects the diagnostic importance of each brain region within the structural connectivity networks of the ADHD group (**a**) and the healthy control (HC) group (**b**), respectively. The color bar indicates the magnitude of average connectivity strength: the closer the color is to red, the higher the value; the closer it is to blue, the lower the value.

46th in the left and right hemispheres, respectively, compared to lower rankings in the HC group (85th and 78th). In contrast, the caudate nucleus and thalamus ranked among the bottom five regions in the ADHD group but ranked between 31st and 38th in the HC group. ACS rankings across all 116 brain regions in the SCNs of both groups are shown in Supplementary Data 1.

Discussion

This study presents an approach that leverages Transformer-based methods to construct personalized SCNs from high-resolution T1WI images for intelligent diagnosis of ADHD. Four image processing modes were proposed to transform brain region images into input for the Transformer encoder, enabling the investigation of interrelationships between brain regions and the establishment of diagnosis models. Results demonstrate that the feature-based Transformer method enhances AUCs to 0.75–0.78 in five-fold cross validation. Through the SCNs constructed during model training, this method effectively captures structural connectivity alterations between ADHD and healthy control (HC) groups, suggesting that ADHD may involve coordinated abnormalities across multiple brain regions. Additionally, SCN analysis revealed significant differences in ACSs between the ADHD and HC groups (paired t-test: $P = 0.81 \times 10^{-6}$, Cohen's d estimate 0.40 [0.24–0.56]). Notably, the bilateral precentral gyrus, left middle frontal gyrus, right triangular part of the inferior frontal gyrus, and bilateral postcentral gyrus were among the top ten most important brain regions in both groups, yet their ACSs varied significantly (paired t-test: $P = 0.13 \times 10^{-4}$, Cohen's d estimate 7.78 [2.09–13.48]). This is the first study to construct

Table. 2 | Top ten important brain regions of the two groups ranked by average connectivity strength (ACS)

ACS ranking (strong to weak)	ADHD group		HC group		Independent *t*-test
	Brain region	ACS	Brain region	ACS	
1	Frontal_Inf_Tri_R	0.66	Postcentral_R	0.51	$P = 0.82 \times 10^{-13}$, Cohen's d estimate 10.20 [6.68–13.71]
2	Precentral_R	0.64	Precentral_R	0.51	
3	Postcentral_R	0.64	Postcentral_L	0.50	
4	Parietal_Sup_R	0.63	Frontal_Inf_Tri_R	0.50	
5	Temporal_Pole_Sup_R	0.62	Precentral_L	0.50	
6	Precentral_L	0.62	Frontal_Mid_L	0.50	
7	Postcentral_L	0.62	Frontal_Inf_Tri_L	0.50	
8	Vermis_3	0.62	Frontal_Inf_Oper_L	0.49	
9	Parietal_Sup_L	0.62	Cerebelum_10_R	0.48	
10	Frontal_Mid_L	0.61	Frontal_Inf_Oper_R	0.48	

ADHD attention-deficit hyperactivity disorder, *HC* healthy control.

personalized, learnable SCNs from high-resolution T1WI images using deep learning. By pioneering a data-driven approach to whole-brain structural connectivity, this study represents a substantial shift from traditional static models toward individualized, dynamic network representations. The proposed method not only enables personalized brain network modeling but also provides promising neuroimaging biomarkers and insights for the clinical diagnosis of ADHD.

A comparative analysis was conducted to evaluate the diagnostic performance of the proposed models against previous T1WI-based deep learning approaches on the ADHD-200 dataset (Supplementary Table 1). Wang et al.[13]. reported a highest accuracy of 78.8%, but the sensitivity was relatively low at 38.1%, highlighting a common issue in existing ADHD diagnostic models. This low sensitivity reflects a clinical challenge, as ADHD symptoms are often misinterpreted as emotional problems. Consequently, some children with ADHD may remain undiagnosed and untreated, resulting in potentially irreversible impacts on their learning and daily lives. Although the accuracy of our F-T model was only 71.9%, it achieved a markedly higher sensitivity of 63.6%, representing a 21.6% improvement over the highest previously reported sensitivity of 42.0%. Moreover, when considering balanced accuracy, which accounts for both sensitivity and specificity, the proposed model outperformed previous studies.

Through the SCNs constructed in this study, we discovered that ADHD may be associated with structural connectivity alterations across multiple brain regions. Traditional T1WI-based studies have hypothesized that the effects of ADHD on the brain manifests as localized structural abnormalities in specific regions or global structural alterations[26,27]. Zhao et al.[28,29]. introduced reginal radiomics similarity networks (R2SNs) by calculating Pearson's correlation coefficient between interregional radiomics features, and used the proposed R2SNs to reveal abnormality patterns in mild cognitive impairment[28,29]. More recently, Liu et al.[30]. also refined the R2SNs and proposed individualized radiomics-based structural similarity network (iRSSN). Differ from these studies that can only compute fixed interregional relationships using Pearson's correlation coefficient, our study is the first to explore the learnable SCNs on T1WI images using Transformer-based framework. In our study, the interregional relationships were not pre-defined but were learned and optimized during training for disease diagnosis. The model assigns different weights to edges between pairs of brain regions, enabling the construction of disease-specific, individualized SCNs.

The Transformer-SCN method introduces a higher computational complexity compared to traditional CNN or graph-based SCN approaches due to the self-attention mechanism, which has a complexity of $O(L^2)$, where L represents the sequence length of brain regions. In contrast, traditional approaches, such as tractography ($O(N)$ or $O(N \log N)$) or CNN-based models ($O(N)$), are less computationally demanding. However, while the computational cost is higher (Supplementary Table 2), the inference time remains acceptable, ranging from 0.0051 s (F-T) to 0.1373 s (VRS-T) per sample. Unlike conventional SCN pipelines[28–30] that require manual SCN computation, our approach directly constructs the SCN within the model, simplifying the inference process and reducing complexity. Although the increased computational cost is a trade-off, the method offers enhanced interpretability and accuracy in ADHD-related structural connectivity analysis, justifying the additional resources required.

The analysis of the SCNs identified key brain regions for the discrimination between the ADHD and HC groups. These regions include the precentral gyrus, left middle frontal gyrus, right triangular part of the inferior frontal gyrus, and bilateral postcentral gyrus, all of which exhibited significantly different ACS between the two groups. Structural and functional alterations in these regions have been linked to ADHD symptoms, including attention difficulties, impulsivity, and irritability[31–33]. Notably, this study also underscores the importance of the cerebellum in ADHD diagnosis, with the cerebellar vermis ranking prominently in the ADHD group and the right inferior cerebellum in the HC group. Although primarily implicated in motor coordination, the cerebellum also influences attention and learning processes. Structural changes, such as reduced volume, have been observed in the cerebellum of individuals with ADHD[34–36], supporting its role in the disorder.

In the ADHD group, we observed an increase in ACS in multiple brain regions, such as the lingual gyrus, precuneus lobe, postcentral gyrus, precentral gyrus, and superior parietal gyrus. These increases may reflect compensatory structural reorganization, where certain regions show stronger morphometric correlations due to neurodevelopmental alterations[37]. ADHD-related structural abnormalities could result in non-canonical coordination patterns across brain regions, contributing to increased connectivity strengths in specific circuits. In contrast, regions such as the caudate nucleus and thalamus exhibited lower ACS in the ADHD group compared to the HC group. These regions, part of the cortico-basal ganglia-thalamic network, are involved in motor, emotional, language, and reward functions. Structural and gene expression abnormalities in the caudate nucleus and thalamus have been reported in ADHD patients[38,39], suggesting that dysfunctions in these areas may contribute to ADHD symptoms such as inattention and impulsivity.

Our findings highlight that ADHD involves complex interactions across multiple brain regions. The T1WI-Transformer models offer a comprehensive method for exploring disease-related structural connectivity abnormalities, providing valuable insights into the neurological mechanisms underlying ADHD.

Several limitations are worth noting in this study. Firstly, the ADHD-200 dataset comprises only 776 samples in the training set, which is relatively small for Transformer models. Expanding the dataset size could potentially enhance the training effectiveness of the models. Secondly, although the fusion models demonstrated improved diagnostic performance over the four models, the improvement was not statistically different when compared to the best-performing F-T model. Future research with larger sample sizes could explore more effective fusion methods to capitalize on the strengths of single models and further enhance diagnostic performance and generalizability. Thirdly, unlike previous studies that manually constructed SCNs using gray matter volume, this study constructed learnable personalized SCNs from 3D T1WI images. This approach enables the detection of abnormalities in both white and gray matter, and facilitates model interpretability[40]. This methodology holds promise for generalization to scenarios involving multi-parameter MRI, multi-modality data, and other brain disease analyses. Incorporating additional MRI sequences for joint analysis based on this method may enhance the elucidation of ADHD-related whole-brain SCN changes. Lastly, several potential confounding variables may influence the observed structural connectivity alterations in ADHD identification. These include age, gender, comorbid psychiatric conditions, medication use, and variability in imaging protocols across sites[41,42]. Such factors are known to affect brain structure and may contribute to differences in connectivity patterns. To assess the impact of comorbidities, we conducted an additional analysis comparing model performance in ADHD participants with and without documented comorbid conditions. Although no significant difference was observed, participants were not fully separated due to incomplete comorbidity records. Future studies will aim to collect larger datasets with detailed profiles to better distinguish ADHD subtypes, co-occurring disorders and other psychiatric disorders. Nonetheless, unmeasured confounders may still exist, and further validation with harmonized imaging and clinical data is warranted to enhance generalizability.

In summary, this study leverages the self-attention mechanism of Transformers to construct brain SCNs from T1WI images, thereby exploring ADHD-related connectivity alterations. The proposed method achieves an AUC of 0.74, a sensitivity of 63.6% and a balanced accuracy of 71.2% on the ADHD-200 testing set, as well as AUCs ranging from 0.75 to 0.78 in five-fold cross validation. SCN analysis revealed significant differences in ACSs between the ADHD and HC groups (paired t-test: $P = 0.81 \times 10^{-6}$, Cohen's d estimate 0.40 [0.24–0.56]), suggesting that ADHD may result from structural connectivity alterations across multiple brain regions. By employing Transformers to construct learnable SCNs, this study represents a departure from traditional static models toward dynamic, data-driven network representations. The proposed method not only enables individualized brain network construction but also provides promising neuroimaging biomarkers and insights for the clinical diagnosis of ADHD.

Data availability

MRI data and clinical characteristics of all participants in this study are publicly accessible through the Neuro Bureau ADHD-200 Preprocessed repository (http://preprocessed-connectomes-project.org/adhd200/download.html). The dataset is publicly available via a public Amazon S3 bucket and the Neuroimaging Informatics Tools and Resources Clearinghouse (NITRC). Data can be downloaded as individual files from S3 using exact URLs or as compressed tar files from NITRC. Users must register for free on NITRC and agree to the data usage policies of the 1000 Functional Connectomes Project and INDI. The dataset is for non-commercial research purposes. The source data underlying the results shown in Table 2, Figs. 2–5, Supplemental Table 1, and Supplemental Fig. 1–11 are available at Supplementary Data 1. All other relevant data are available from the corresponding author upon reasonable request.

Code availability

The codes are available at https://github.com/sekirete/ADHD_Transformer[43].

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Author contributions

Xin Gao and Liting Shi have full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. The study was conceptualized and designed by Liting Shi, Xin Gao, and Lei Shi. Data were acquired, analyzed, or interpreted by Liting Shi, Lei Shi, Zhijun Cui, Rui Zhang, Jiayi Zhang, Wei Shi, Yechen Zhu, Jianlin Wang, Yanlong Wang, and Dongxing Wang. The manuscript was drafted by Liting Shi and Chengting Lin, and all authors critically revised it for important intellectual content. Statistical analysis was performed by Liting Shi and Lei Shi. Funding was obtained by Xin Gao, Rui Zhang, and Jiayi Zhang. Administrative, technical, or material support was provided by Liting Shi, Xin Gao, and Haihong Liu. The study was supervised by Xin Gao and Lei Shi. Liting Shi and Lei Shi contributed equally to this work.

Competing interests

The authors declare no competing interests.

Additional information

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